

PRAJWAL KUMAR K

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LinkedIn | GitHub | Portfolio
Codeforces | CodeChef | LeetCode | AtCoder

EDUCATION

Indian Institute of Information Technology, Kottayam

B.Tech in Computer Science and Engineering; CGPA: 8.66/10.0

Kottayam, Kerala

Aug 2024 – May 2028 (expected)

RESEARCH INTERESTS

Algorithms and systems, game theory and strategic decision-making, machine learning and data science, AI safety, and NLP.

RESEARCH AND SELECTED PROJECTS

PokeForge | *Pokémon Battle Agents, PyTorch*

GitHub | In progress

- Developing a battle-agent prototype; reproduced pinned FoulPlay and Metamon Gen 9 OU baselines in isolated environments, with simulator conformance and competitive strength still under evaluation.

Bluff | *Imperfect-Information Agents, Bayesian Opponent Models*

GitHub

- Built a repeated-play imperfect-information card-game simulator to study agent adaptation against recurring opponents, using persistent Bayesian profiles and a learned bluff detector.
- A seeded ablation across eight synthetic profiles found that removing the detector reduced draw-aware score by 4.9–5.8 percentage points; results are limited to this simulator and roster.

Drishti-XAI | *Retinal-Image Classification and Evaluation*

Team SIH 2026 submission

- Built the MATLAB/Python prototype and evaluation pipeline for a team project, including image-quality checks, grading, segmentation, Grad-CAM explanations, calibration, and reporting.
- Evaluated the same model and threshold on internal APTOS (n=733; sensitivity 0.908, specificity 0.877) and external Messidor-2 (sensitivity 0.363, specificity 0.865, AUC 0.666), documenting a substantial transfer gap.

KPlaceNet | *Image Geolocation, PyTorch, ResNet-50*

GitHub | Team project

- Evaluated CNN geolocation with quadtree cells; the best of 12 ResNet-50 runs placed 3.33% of 3,000 spatially separated OSV-5M test images within 200 km, exposing weak generalization.

PUBLICATION

K. Prajwal Kumar, Rubiya Subair, and Shajulin Benedict. “Collaboration-Enabled Energy-Aware Learning Model Exchange (CE-LME) Framework for Edge Computing-Assisted Applications.” *2025 IEEE 3rd International Conference on Inventive Computing and Informatics*, pp. 1390–1395. doi:10.1109/ICICI65870.2025.11069657. First-listed author; conference presenter.

TECHNICAL SKILLS AND STUDY

Programming: Python, C++, Rust, MATLAB | **ML:** PyTorch, NumPy, scikit-learn

Spoken languages: English, German, Hindi, Kannada

Independent study, in progress: An Introduction to Statistical Learning; Mathematics for Machine Learning; Stanford CS231n.

LEADERSHIP AND RECOGNITION

Technical Lead, Openverse; Core Technical Member, Coders Club, IIIT Kottayam. Algorithmima was selected for OOSC 4.0; the team gave a lightning talk at OOSC 3.0 and presented at T4E 2025, IIT Madras. Best Software Award, IEEE MACE .hack(); '26, Team Hierarchy.